

Elements	Description
Test plan	- This is considered the container of JMeter. It consists of various thread groups, controllers, listeners, configuration elements, and so on. - Every test plan should consist of at least one thread group.
Work bench	- This provides the capability to store the elements temporarily. - Elements from the workbench can be copied and pasted under the thread group as per requirements. - On saving the test plan, elements under the workbench will not be saved by default.
Thread group	This element is used to define the number of threads (users) for which the script will run. It provides the flexibility to: - Set the number of threads - Set the number of iterations - Set the ramp-up time
Listeners	This is used to view the test results. Various types of listeners are available in JMeter. Some of the commonly used ones are: - View results in the table - View the results tree - Graph results - Aggregate report
Assertions	These are used to validate the response received from the application/server under load. Various types of assertions can be added, which depends on the response that needs to be validated. Commonly used assertions are: - HTML assertion - Size assertion - Response assertion - Bean shell assertion - Duration assertion
Timers	These are used to provide a delay between various requests within the same thread group or within various thread groups. Some of the commonly used timers are: - Constant timer - Gaussian timer - Bean shell timer
Test fragment	This is a type of controller on the same level as the thread group. But the difference between the test fragment and the thread group is that it cannot be executed without the module controller and includes the controller. This element is used in distributed testing.
Pre-processors	These are used to modify, create or update the requests or variables. It is executed just prior to when the request is made. Commonly used pre-processors are: - User parameters - Bean shell processors - JDBC processors - Regex user parameters
Post processors	These are used to process the response data, to extract the values from it, and so on. They are executed just after the request is made. Commonly used post processors are: - Regular expression extractor - XPath extractor - DBC PostProcessor
Samplers	These are the requests sent by JMeter to the server. They wait till the response is received. Various types of samplers are available as per the request sent. Commonly used versions are: - HTTP request - JDBC request - FTP request
HTTP cache manager	This provides the caching functionality to HTTP requests in JMeter to stimulate real browser caching. Each user in JMeter will have its own cache.
HTTP cookie manager	- This stores and sends the cookies to the browser. - If any response or HTTP requests consist of cookies, these will get automatically stored in the cookie manager while recording, and can be used in future when this response or requests are run again for that application.
HTTP(s) test script recorder	- This provides JMeter the capability to record and intercept the action when you browse your application in a normal browser. - Test sample objects are created and stored in the test plan. - Port number should match with browser's port number.

Table 1